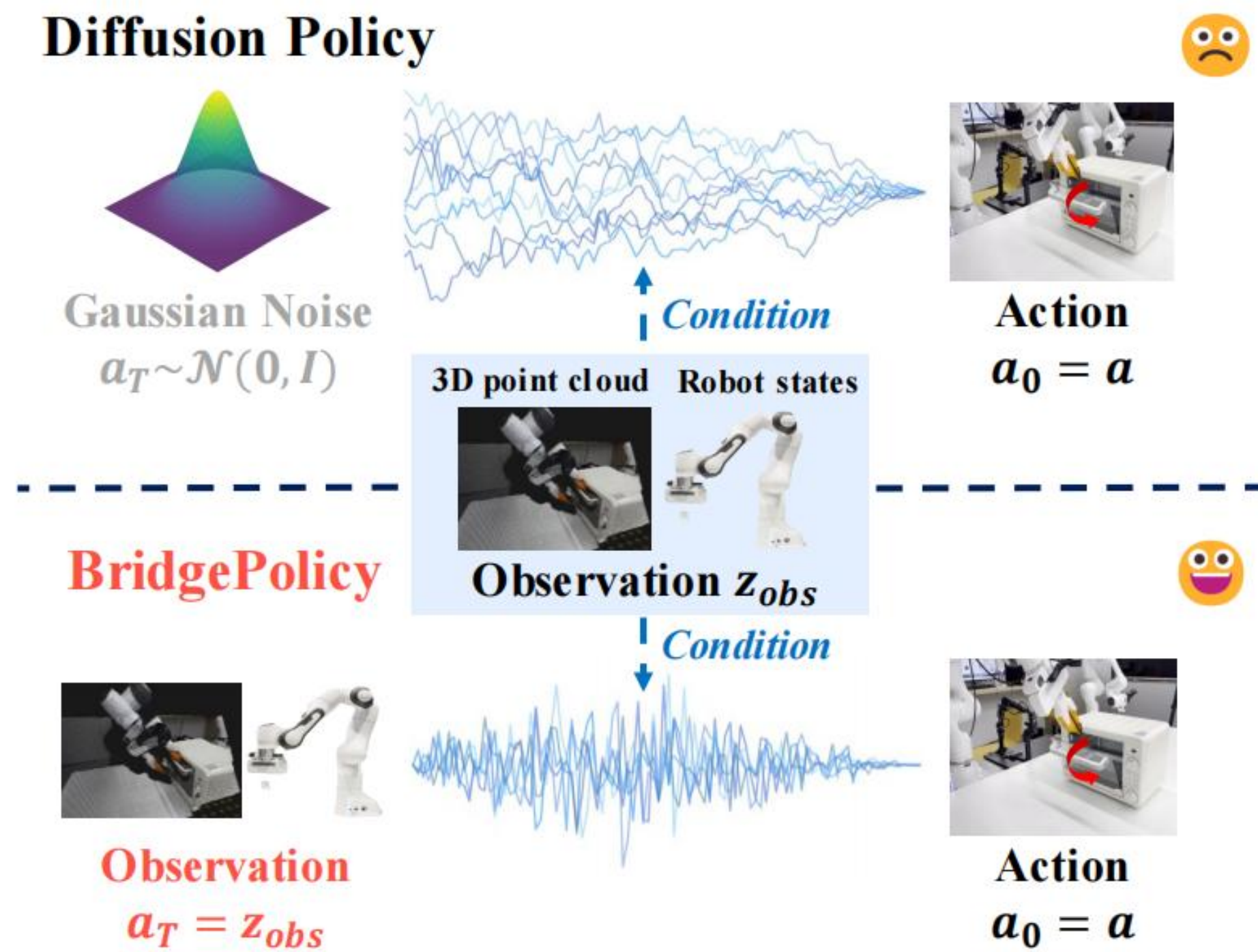


Sample from What You See: Visuomotor Policy Learning via Diffusion Bridge with Observation-Embedded Stochastic Differential Equation

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Motivation:

Traditional diffusion-based policy underutilized the observation information by only treating them as high-level condition during the sampling process, leading to suboptimal performance. 😞



Our Solution:

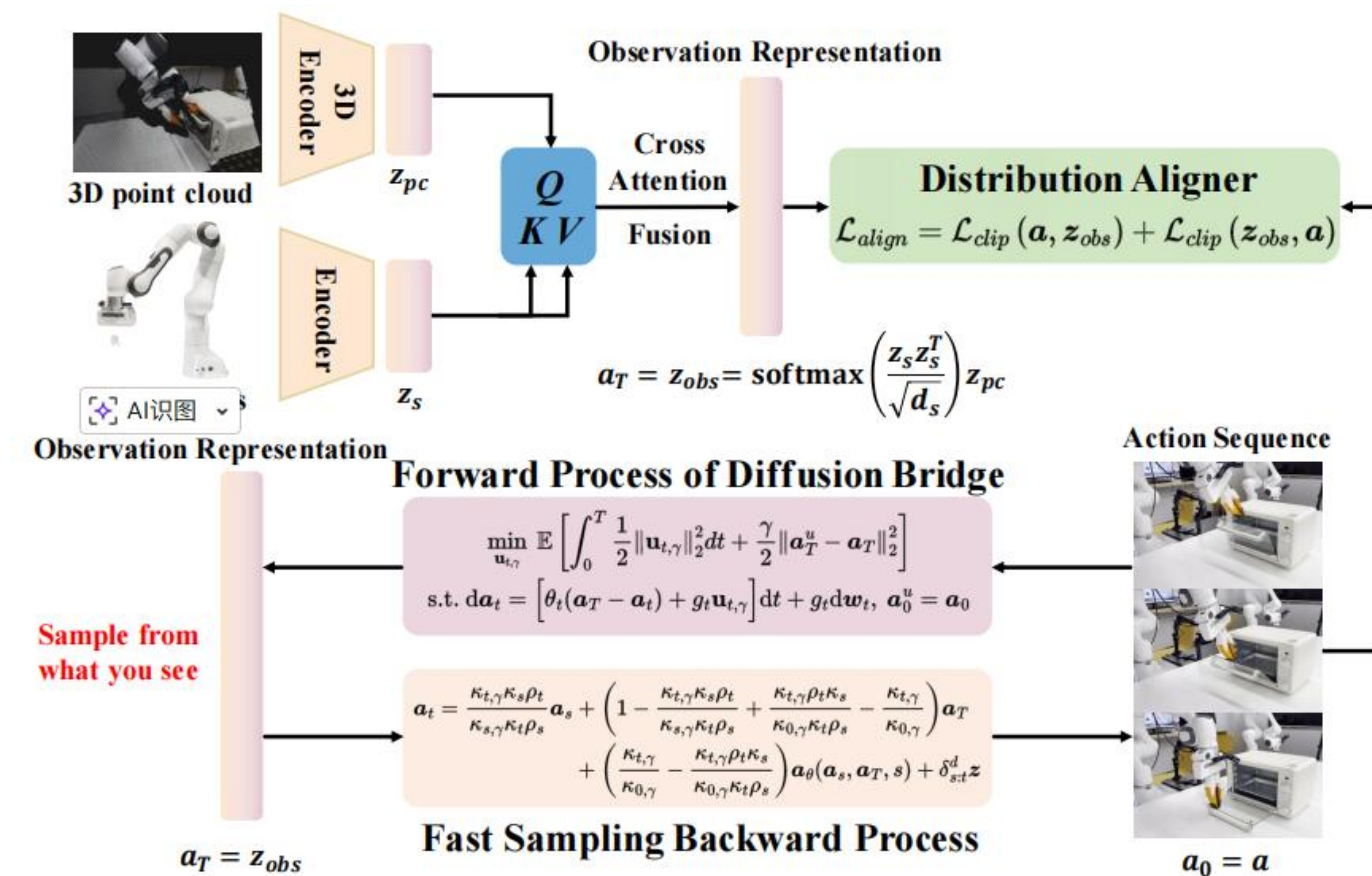
By constructing an observation-informed diffusion SDE trajectory via **Diffusion Bridge**, **BridgePolicy** enables sampling to start from a rich, informative prior rather the uninformative random noise.

Challenges:

- Diffusion Bridge can only connect one-to-one distributions while the observations are inherently multi-modal. 🤔
- Standard Diffusion Bridge require the connected distributions share the same data shape, which is barely satisfied in robotic observations.

Method:

To this end, we propose **BridgePolicy**, a novel generative policy that embed the observation into the diffusion SDE trajectory via the formulation of diffusion bridge, therefore directly samples actions from observation-informed prior rather than random noise.



Simulation tasks

Methods\Task	MetaWorld Easy	MetaWorld Medium	MetaWorld Hard	MetaWorld Very Hard	DexArt	Adroit	Average
DP [5]	0.79±0.27	0.31±0.24	0.10±0.12	0.26±0.25	0.45±0.08	0.31±0.17	0.37
DP3 [55]	0.87±0.20	0.61±0.29	0.40±0.37	0.51±0.33	0.57±0.08	0.68±0.06	0.60
Simple DP3 [55]	0.86±0.12	0.59±0.22	0.38±0.26	0.47±0.28	0.48±0.07	0.68±0.05	0.58
FlowPolicy [56]	0.86±0.16	0.67±0.21	0.59±0.30	0.76±0.26	0.54±0.09	0.70±0.12	0.68
BridgePolicy (Ours)	0.91±0.24	0.75±0.28	0.58±0.31	0.79±0.32	0.60±0.07	0.81±0.12	0.74

Quantitative Results:

Achieve state-of-the-art performance on 52 simulation task across 3 benchmarks and 5 real-world tasks.

Real-world tasks

	Oven-Closing	Oven-Opening	Pick Place	Pour	Unplug	Average
Simple DP3	0.8	0.6	0.6	0.6	0.7	0.66
DP3	0.9	0.9	0.7	0.6	0.7	0.76
FlowPolicy	1.0	0.7	0.5	0.1	0.5	0.56
BridgePolicy	1.0	1.0	0.8	0.8	0.9	0.90

Qualitative Results:

Demonstrate more stable and controllable behaviors than the Diffusion-based and Flow-based policies when performing the real-world tasks.

